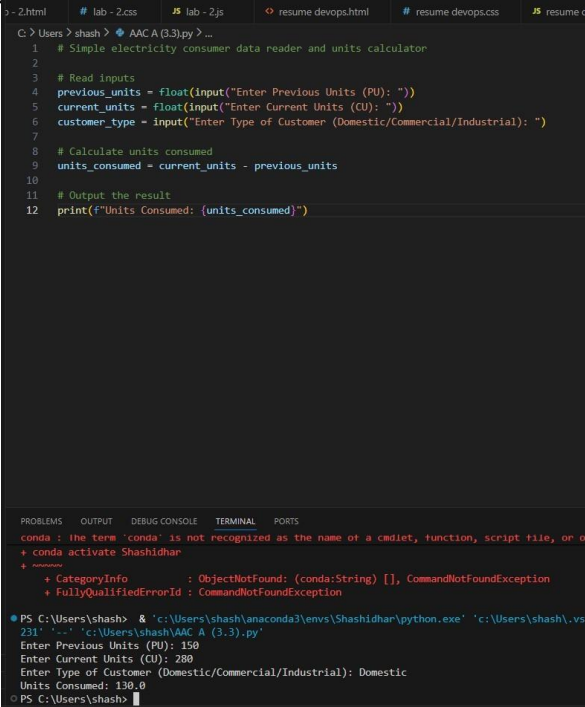


SCHOOL OF COMPUTER SCIENCE AND ARTIFICIAL INTELLIGENCE		DEPARTMENT OF COMPUTER SCIENCE ENGINEERING	
Program Name: B. Tech		Assignment Type: Lab	Academic Year: 2025-2026
Course Coordinator Name		Dr. Rishabh Mittal	
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		Ms. M.Madhuri	
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		Mr. Bingi Raju	
Course Code	23CS002PC304	Course Title	AI Assisted Coding
Year/Sem	III/I	Regulation	R23
Date and Day of Assignment	Week 2 - Wednesday	Time(s)	23CSBTB01 To 23CSBTB52
Duration	2 Hours	Applicable to Batches	All batches
Assignment Number: 3.3(Present assignment number)/24(Total number of assignments)			
Q.No.	Question	Expected Time to complete	
1	Lab 3: Application for TGNPDCL – Electricity Bill Generation Using Python & AI Tools Lab Objectives - To design a real-world electricity billing application using Python - To use AI-assisted coding tools for logic generation and optimization - To understand conditional logic and arithmetic operations - To generate structured billing output similar to utility bills Lab Outcomes (LOs) After completing this lab, students will be able to:	Week2 - Wednesday	

	• Read and validate user input in Python • Apply conditional logic for tariff-based billing • Use AI tools to assist in program development • Calculate and display electricity bill components • Build a complete real-time application Task 1: AI-Generated Logic for Reading Consumer Details Scenario An electricity billing system must collect accurate consumer data. Task Description Use an AI tool (GitHub Copilot / Gemini) to generate a Python program that: • Reads: ○ Previous Units (PU) ○ Current Units (CU) ○ Type of Customer • Calculates units consumed • Implements logic directly in the main program (no functions) Expected Output  <pre> C:\Users\shash> python AACA (3.3).py 1 # Simple electricity consumer data reader and units calculator 2 3 # Read inputs 4 previous_units = float(input("Enter Previous Units (PU): ")) 5 current_units = float(input("Enter Current Units (CU): ")) 6 customer_type = input("Enter Type of Customer (Domestic/Commercial/Industrial): ") 7 8 # Calculate units consumed 9 units_consumed = current_units - previous_units 10 11 # Output the result 12 print(f"Units Consumed: {units_consumed}") </pre> <pre> PS C:\Users\shash> & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\vs 231' '-.' 'c:\Users\shash\AACA (3.3).py' Enter Previous Units (PU): 150 Enter Current Units (CU): 280 Enter Type of Customer (Domestic/Commercial/Industrial): Domestic Units Consumed: 130.0 PS C:\Users\shash> </pre> Task 2: Energy Charges Calculation Based on Units Consumed Scenario Energy charges depend on the number of units consumed and customer type. Task Description Review the AI-generated code from Task 1 and extend it to: • Calculate Energy Charges (EC) • Use conditional statements based on: ○ Domestic ○ Commercial ○ Industrial consumers • Improve readability using AI prompts such as:
--	---

Expected

-
-
-
-

```
○ "Simplify energy charge calculation logic"
○ "Optimize conditional statements"

C:\Users\shash > cd AAC A (3.3).py > ...
4 previous_units = float(input("Enter Previous Units (PU): "))
5 current_units = float(input("Enter Current Units (CU): "))
6 customer_type = input("Enter Type of Customer (Domestic/Commercial/Industrial): ")
7
8 # Calculate units consumed
9 units_consumed = current_units - previous_units
10
11 # Calculate Energy Charges (EC) based on type and slabs
12 if customer_type == "Domestic":
13     if units_consumed <= 100:
14         ec = units_consumed * 1.0
15     elif units_consumed <= 200:
16         ec = 100 * 1.0 + (units_consumed - 100) * 2.0
17     else:
18         ec = 100 * 1.0 + 100 * 2.0 + (units_consumed - 200) * 3.0
19 elif customer_type == "Commercial":
20     if units_consumed <= 100:
21         ec = units_consumed * 1.5
22     elif units_consumed <= 200:
23         ec = 100 * 1.5 + (units_consumed - 100) * 2.5
24     else:
25         ec = 100 * 1.5 + 100 * 2.5 + (units_consumed - 200) * 4.0
26 elif customer_type == "Industrial":
27     if units_consumed <= 100:
28         ec = units_consumed * 2.0
29     elif units_consumed <= 200:
30         ec = 100 * 2.0 + (units_consumed - 100) * 3.0
31     else:
32         ec = 100 * 2.0 + 100 * 3.0 + (units_consumed - 200) * 5.0
33 else:
34     ec = 0 # Invalid type
35     print("Invalid customer type!")
36
37 # Output
38 print(f"Units Consumed: {units_consumed}")
39 print(f"Energy Charges (EC): ${ec:.2f}")

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

Enter Previous Units (PU): 150
Enter Current Units (CU): 280
Enter Type of Customer (Domestic/Commercial/Industrial): Domestic
Units Consumed: 130.0
PS C:\Users\shash> cd 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\AAC A (3.3).py'
Enter Previous Units (PU): 150
Enter Current Units (CU): 280
Enter Type of Customer (Domestic/Commercial/Industrial): Domestic
Units Consumed: 130.0
Energy Charges (EC): $160.00
PS C:\Users\shash>
```

Task 3: Modular Design Using AI Assistance (Using Functions)
Scenario

Billing logic must be reusable for multiple consumers.

Task Description

Use AI assistance to generate a Python program that:

- Uses user-defined functions to:
 - Calculate Energy Charges
 - Calculate Fixed Charges
- Returns calculated values
- Includes meaningful comments

Expected Output

- Function-based Python program
- Correct EC and FC values
- Screenshots of AI-assisted function generation
- Test cases with outputs

```

C:\Users> shash > AACA (3.3).py > ...
1 # Modular Electricity Billing System
2
3 def calculate_energy_charges(customer_type, units_consumed):
4     """
5     Calculate Energy Charges based on customer type and units consumed.
6     Slabs: Domestic (1/2/3), Commercial (1.5/2.5/4), Industrial (2/3/5) per unit tiers.
7     """
8     if customer_type == "Domestic":
9         if units_consumed <= 100:
10             return units_consumed * 1.0
11         elif units_consumed <= 200:
12             return 100 * 1.0 + (units_consumed - 100) * 2.0
13         else:
14             return 100 * 1.0 + 100 * 2.0 + (units_consumed - 200) * 3.0
15     elif customer_type == "Commercial":
16         if units_consumed <= 100:
17             return units_consumed * 1.5
18         elif units_consumed <= 200:
19             return 100 * 1.5 + (units_consumed - 100) * 2.5
20         else:
21             return 100 * 1.5 + 100 * 2.5 + (units_consumed - 200) * 4.0
22     elif customer_type == "Industrial":
23         if units_consumed <= 100:
24             return units_consumed * 2.0
25         elif units_consumed <= 200:
26             return 100 * 2.0 + (units_consumed - 100) * 3.0
27         else:
28             return 100 * 2.0 + 100 * 3.0 + (units_consumed - 200) * 5.0
29     return 0 # Invalid type
30
31 def calculate_fixed_charges(customer_type):
32     """
33     Calculate Fixed Charges based on customer type.
34     Domestic: $100, Commercial: $200, Industrial: $300.
35     """
36     if customer_type == "Domestic":
37         return 100.0
38     elif customer_type == "Commercial":
39         return 200.0
40     elif customer_type == "Industrial":
41         return 300.0
42     return 0 # Invalid type
43
44 # Main program
45 previous_units = float(input("Enter Previous Units (PU): "))
46 current_units = float(input("Enter Current Units (CU): "))
47 customer_type = input("Enter Type of Customer (Domestic/Commercial/Industrial): ")
48
49 units_consumed = current_units - previous_units
50 ec = calculate_energy_charges(customer_type, units_consumed)
51 fc = calculate_fixed_charges(customer_type)
52
53 print(f"Units Consumed: {units_consumed}")
54 print(f"Energy Charges (EC): ${ec:.2f}")
55 print(f"Fixed Charges (FC): ${fc:.2f}")

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

sh\AACA (3.3).py
Units Consumed: 130.0
Energy Charges (EC): \$160.00
PS C:\Users\shash> cd 'c:\Users\shash' & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\AACA (3.3).py'
Enter Previous Units (PU): 150
Enter Current Units (CU): 280
Enter Type of Customer (Domestic/Commercial/Industrial): Domestic
Units Consumed: 130.0
Energy Charges (EC): \$160.00
Fixed Charges (FC): \$100.00
PS C:\Users\shash>

Welcome SubSetSum.java lab - 2.html # lab - 2.css JS lab - 2.js resume dev

```

C:\Users> shash > AACA (3.3).py > ...
25     elif units_consumed <= 200:
26         return 100 * 2.0 + (units_consumed - 100) * 3.0
27     else:
28         return 100 * 2.0 + 100 * 3.0 + (units_consumed - 200) * 5.0
29     return 0 # Invalid type
30
31 def calculate_fixed_charges(customer_type):
32     """
33     Calculate Fixed Charges based on customer type.
34     Domestic: $100, Commercial: $200, Industrial: $300.
35     """
36     if customer_type == "Domestic":
37         return 100.0
38     elif customer_type == "Commercial":
39         return 200.0
40     elif customer_type == "Industrial":
41         return 300.0
42     return 0 # Invalid type
43
44 # Main program
45 previous_units = float(input("Enter Previous Units (PU): "))
46 current_units = float(input("Enter Current Units (CU): "))
47 customer_type = input("Enter Type of Customer (Domestic/Commercial/Industrial): ")
48
49 units_consumed = current_units - previous_units
50 ec = calculate_energy_charges(customer_type, units_consumed)
51 fc = calculate_fixed_charges(customer_type)
52
53 print(f"Units Consumed: {units_consumed}")
54 print(f"Energy Charges (EC): ${ec:.2f}")
55 print(f"Fixed Charges (FC): ${fc:.2f}")

```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\shash> cd 'c:\Users\shash' & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\AACA (3.3).py'
Fixed Charges (FC): \$100.00
PS C:\Users\shash> cd 'c:\Users\shash' & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\AACA (3.3).py'
Enter Previous Units (PU): 0
Enter Current Units (CU): 250
Enter Type of Customer (Domestic/Commercial/Industrial): Commercial
Units Consumed: 250.0
Energy Charges (EC): \$600.00
Fixed Charges (FC): \$200.00
PS C:\Users\shash>

Scenario

Extend the program to calculate:

- **FC** – Fixed Charges
- **CC** – Customer Charges
- **ED** – Electricity Duty (percentage of EC)

- *"Add electricity duty calculation"*
- *"Improve billing accuracy"*

•
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```

1 # Extended Electricity Billing with Additional Charges
2
3 def calculate_energy_charges(customer_type, units_consumed):
4     """
5     Calculate Energy Charges based on customer type and units consumed.
6     Slabs: Domestic (1/2/3), Commercial (1.5/2.5/4), Industrial (2/3/5) per unit tiers.
7     """
8     if customer_type == "Domestic":
9         if units_consumed <= 100:
10             return units_consumed * 1.0
11         elif units_consumed <= 200:
12             return 100 * 1.0 + (units_consumed - 100) * 2.0
13         else:
14             return 100 * 1.0 + 100 * 2.0 + (units_consumed - 200) * 3.0
15     elif customer_type == "Commercial":
16         if units_consumed <= 100:
17             return units_consumed * 1.5
18         elif units_consumed <= 200:
19             return 100 * 1.5 + (units_consumed - 100) * 2.5
20         else:
21             return 100 * 1.5 + 100 * 2.5 + (units_consumed - 200) * 4.0
22     elif customer_type == "Industrial":
23         if units_consumed <= 100:
24             return units_consumed * 2.0
25         elif units_consumed <= 200:
26             return 100 * 2.0 + (units_consumed - 100) * 3.0
27         else:
28             return 100 * 2.0 + 100 * 3.0 + (units_consumed - 200) * 5.0
29     return 0 # Invalid type
30
31 def calculate_fixed_charges(customer_type):
32     """
33     Calculate Fixed Charges based on customer type.
34     Domestic: $100, Commercial: $200, Industrial: $300.
35     """
36     if customer_type == "Domestic":
37         return 100.0
38     elif customer_type == "Commercial":
39         return 200.0
40     elif customer_type == "Industrial":
41         return 300.0
42     return 0 # Invalid type
43
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```

```

C:\Users\shash > AAC A (3.3).py > ...
31 def calculate_fixed_charges(customer_type):
32     return 100.0
33 elif customer_type == "Commercial":
34     return 200.0
35 elif customer_type == "Industrial":
36     return 300.0
37 return 0 # Invalid type
38
39 def calculate_customer_charges():
40     """Fixed Customer Charges: $50 for all types."""
41     return 50.0
42
43 def calculate_electricity_duty(ec):
44     """Electricity Duty: 10% of Energy Charges."""
45     return 0.10 * ec
46
47 # Main program
48 previous_units = float(input("Enter Previous Units (PU): "))
49 current_units = float(input("Enter Current Units (CU): "))
50 customer_type = input("Enter Type of Customer (Domestic/Commercial/Industrial): ").strip()
51
52 if customer_type not in ["Domestic", "Commercial", "Industrial"]:
53     print("Invalid type! Defaulting to Domestic.")
54     customer_type = "Domestic"
55
56 units_consumed = current_units - previous_units
57 ec = calculate_energy_charges(customer_type, units_consumed)
58 fc = calculate_fixed_charges(customer_type)
59 cc = calculate_customer_charges()
60 ed = calculate_electricity_duty(ec)
61
62 # Print individual charges
63 print(f"Units Consumed: {units_consumed}")
64 print(f"Energy Charges (EC): ${ec:.2f}")
65 print(f"Fixed Charges (FC): ${fc:.2f}")
66 print(f"Customer Charges (CC): ${cc:.2f}")
67 print(f"Electricity Duty (ED): ${ed:.2f}")
68
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
conda : The term 'conda' is not recognized as the name of a cmdlet, function, script file, or operable pro
+ FullyQualifiedErrorId : CommandNotFoundException
PS C:\Users\shash> & 'C:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'C:\Users\shash\.vscode\extens
Enter Previous Units (PU): 150
Enter Current Units (CU): 280
Enter Type of Customer (Domestic/Commercial/Industrial): Domestic
Units Consumed: 130.0
Energy Charges (EC): $160.00
Fixed Charges (FC): $100.00
Customer Charges (CC): $50.00
Electricity Duty (ED): $16.00
PS C:\Users\shash>

```

Task 5: Final Bill Generation and Output Analysis
Scenario

The final electricity bill must present all values clearly.

Task Description

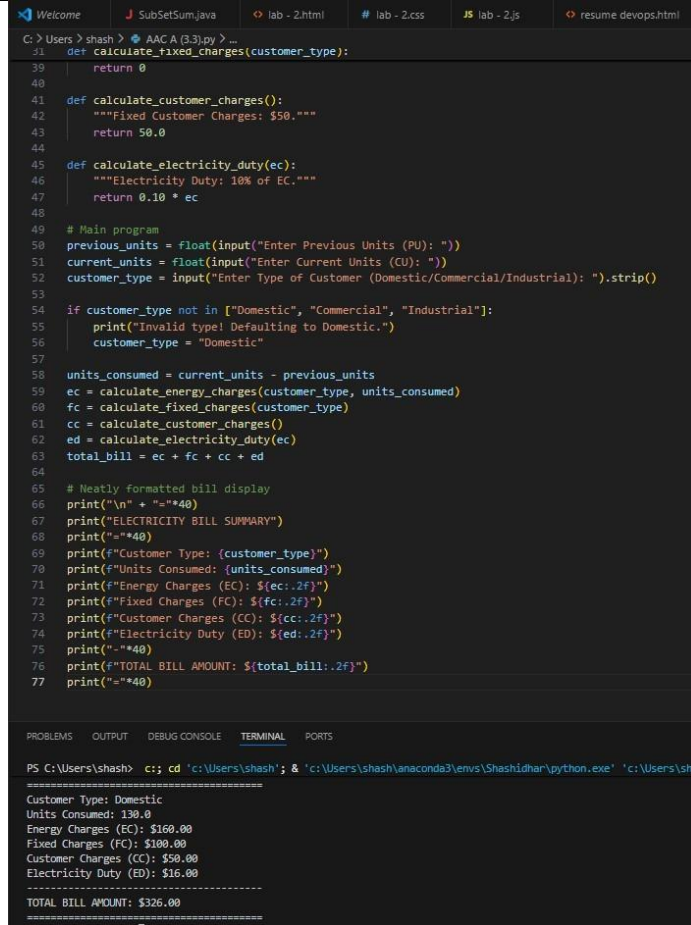
Develop the final Python application to:

- Calculate total bill:
- Total Bill = EC + FC + CC + ED
- Display:
 - Energy Charges (EC)
 - Fixed Charges (FC)
 - Customer Charges (CC)
 - Electricity Duty (ED)
 - Total Bill Amount
- Analyze the program based on:
 - Accuracy
 - Readability
 - Real-world applicability

Expected Output

- Complete electricity bill output
- Neatly formatted display
- Sample input/output
- Short analysis paragraph

```
C:\Users\shash > shash > AAC A (3.3).py > ...
1 # Final Electricity Bill Generator
2
3 def calculate_energy_charges(customer_type, units_consumed):
4     """
5     Calculate Energy Charges based on customer type and units consumed.
6     Slabs: Domestic (1/2/3), Commercial (1.5/2.5/4), Industrial (2/3/5) per unit tiers.
7     """
8     if customer_type == "Domestic":
9         if units_consumed <= 100:
10             return units_consumed * 1.0
11         elif units_consumed <= 200:
12             return 100 * 1.0 + (units_consumed - 100) * 2.0
13         else:
14             return 100 * 1.0 + 100 * 2.0 + (units_consumed - 200) * 3.0
15     elif customer_type == "Commercial":
16         if units_consumed <= 100:
17             return units_consumed * 1.5
18         elif units_consumed <= 200:
19             return 100 * 1.5 + (units_consumed - 100) * 2.5
20         else:
21             return 100 * 1.5 + 100 * 2.5 + (units_consumed - 200) * 4.0
22     elif customer_type == "Industrial":
23         if units_consumed <= 100:
24             return units_consumed * 2.0
25         elif units_consumed <= 200:
26             return 100 * 2.0 + (units_consumed - 100) * 3.0
27         else:
28             return 100 * 2.0 + 100 * 3.0 + (units_consumed - 200) * 5.0
29     return 0
30
31 def calculate_fixed_charges(customer_type):
32     """Fixed Charges: Domestic $100, Commercial $200, Industrial $300."""
33     if customer_type == "Domestic":
34         return 100.0
35     elif customer_type == "Commercial":
36         return 200.0
37     elif customer_type == "Industrial":
38         return 300.0
39     return 0
40
41 def calculate_customer_charges():
42     """Fixed Customer Charges: $50 """
43
44 PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
45
46 PS C:\Users\shash> c:; cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\V
47 Enter Previous Units (PU): 150
48 Enter Current Units (CU): 200
49 Enter Type of Customer (Domestic/Commercial/Industrial): Domestic
50
51 =====
52 ELECTRICITY BILL SUMMARY
53 =====
54 Customer Type: Domestic
55 Units Consumed: 130.0
56 Energy Charges (EC): $160.00
57 Fixed Charges (FC): $100.00
```

```
C:\> Users > shash > AAC A (3.3).py > ...
31 def calculate_fixed_charges(customer_type):
32     return 0
33
34
35
36
37
38
39 return 0
40
41 def calculate_customer_charges():
42     """Fixed Customer Charges: $50."""
43     return 50.0
44
45
46
47
48
49
50 def calculate_electricity_duty(ec):
51     """Electricity Duty: 10% of EC."""
52     return 0.10 * ec
53
54
55
56
57
58
59 # Main program
60 previous_units = float(input("Enter Previous Units (PU): "))
61 current_units = float(input("Enter Current Units (CU): "))
62 customer_type = input("Enter Type of Customer (Domestic/Commercial/Industrial): ").strip()
63
64
65
66
67
68
69 if customer_type not in ["Domestic", "Commercial", "Industrial"]:
70     print("Invalid type! Defaulting to Domestic.")
71     customer_type = "Domestic"
72
73
74
75
76
77
78 units_consumed = current_units - previous_units
79 ec = calculate_energy_charges(customer_type, units_consumed)
80 fc = calculate_fixed_charges(customer_type)
81 cc = calculate_customer_charges()
82 ed = calculate_electricity_duty(ec)
83 total_bill = ec + fc + cc + ed
84
85
86
87
88
89
90 # Neatly formatted bill display
91 print("\n" + "="*40)
92 print("ELECTRICITY BILL SUMMARY")
93 print("="*40)
94 print(f"Customer Type: {customer_type}")
95 print(f"Units Consumed: {units_consumed}")
96 print(f"Energy Charges (EC): ${ec:.2f}")
97 print(f"Fixed Charges (FC): ${fc:.2f}")
98 print(f"Customer Charges (CC): ${cc:.2f}")
99 print(f"Electricity Duty (ED): ${ed:.2f}")
100 print("-"*40)
101 print(f"TOTAL BILL AMOUNT: ${total_bill:.2f}")
102 print("="*40)

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\shash> c: cd 'c:\Users\shash'; & 'c:\Users\shash\anaconda3\envs\Shashidhar\python.exe' 'c:\Users\shash\AAC A (3.3).py'
=====
Customer Type: Domestic
Units Consumed: 130.0
Energy Charges (EC): $160.00
Fixed Charges (FC): $100.00
Customer Charges (CC): $50.00
Electricity Duty (ED): $16.00
=====
TOTAL BILL AMOUNT: $326.00
=====
```

Note: Report should be submitted as a word document for all tasks in a single document with prompts, comments & code explanation, and output and if required, screenshots.